THE NEUROFIBROMATOSES

AT A GLANCE

Neurofibromatosis type 1 (NF-1), an autosomal dominant condition, occurs at an incidence of 1 in 3000 live births. Neurofibromatosis type 2 (NF-2), a separate autosomal dominant disorder, occurs at an incidence of 1 in 40,000 live births.

NF-1 can be diagnosed clinically if two of the following features are present:
- Six or more café-au-lait spots (larger than 0.5 cm in children, larger than 1.5 cm in adults)
- Intertriginous freckling
- Two or more cutaneous neurofibromas or one plexiform neurofibroma
- Optic nerve glioma
- Characteristic bony lesion
- Iris Lisch nodules
- A first-degree relative with NF-1

Cutaneous neurofibromas are softer than the surrounding connective tissue and protrude just above the skin surface or lie just under the skin. Subcutaneous neurofibromas, which arise from peripheral nerves and can be located under the skin or deep in the viscera, are generally much harder.

Plexiform neurofibromas, typically present at birth or evident within the first few years of life, may cause disfigurement, blindness (due to amblyopia, glaucoma, or proptosis), loss of limb function, or organ dysfunction through compression of vital structures. These individuals appear to have an increased risk of facial dysmorphism, mental retardation, early appearance of neurofibromas, and the presence of plexiform neurofibromas. There is evidence that NF-1 is more severe when inherited from the mother rather than the father.

The hallmark of NF-2 is the presence of bilateral vestibular schwannomas. Individuals with NF-2 are also at increased risk for developing multiple meningiomas, schwannomas, gliomas, and neurofibromas throughout the neural axis.

Segmental NF-1 involves manifestations of NF-1—typically café-au-lait macules and neurofibromas—limited to one region of the body. This results from a post-zygotic mutation in the *NF1* gene, leading to somatic mosaicism.

Molecular Biology

The *NF1* gene is located on the long arm of chromosome 17 and encodes a protein called **neurofibromin**. Homozygosity for mutant *NF1* alleles has not been reported, likely because it is a lethal condition.

NF1 has an unusually high mutation rate, estimated at 2.4 to 10×10^{-5} gametes per generation—one of the highest rates among known inherited disorders. This may be due to the large size of the gene, which spans $350\ \text{kb}$ of genomic DNA and contains 59 exons encoding a protein of over 2800 amino acids. Alternatively, the gene may have structural features that make it particularly prone to mutation.

New mutations account for approximately 50% of NF-1 cases and occur most frequently on the paternally inherited allele. Unlike many other autosomal dominant disorders, the frequency of new mutations does not increase with advancing paternal age.

Most *NF1* mutations lead to reduced intracellular levels of neurofibromin, which is sufficient to cause the clinical features of the disease. Tumorigenesis—such as the development of benign dermal neurofibromas—requires inactivation of the normal *NF1* allele in somatic cells, a process known as **loss of heterozygosity**. This two-hit mechanism is characteristic of **tumor suppressor genes**, as seen in other inherited cancer predisposition syndromes like retinoblastoma and Li-Fraumeni syndrome. Tumor formation occurs when both copies of the gene are inactivated: the first by an inherited mutation, the second by a somatic mutation.